Get started with prompt flow to develop language model apps in the Microsoft Foundry

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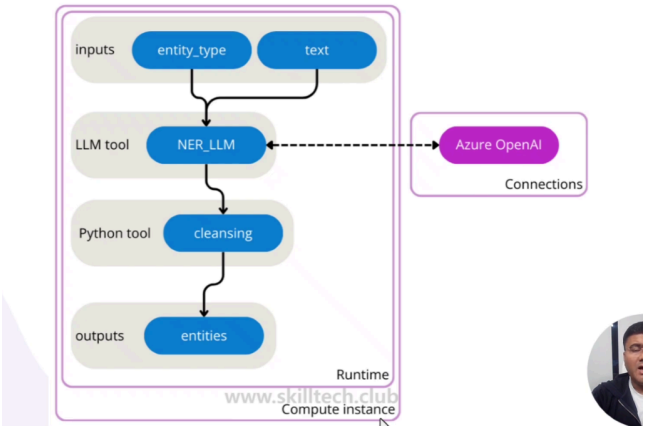
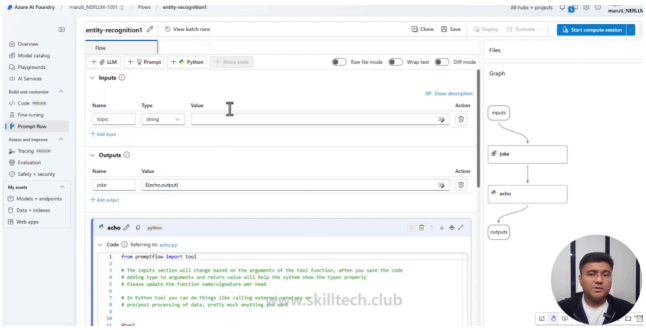
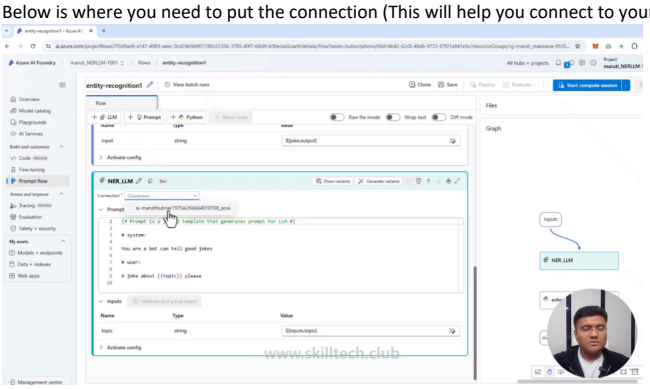
1. **Introduction**
   a. Prompt flow takes a prompt as input, which in the context of LLMs, refers to the query provided to the LLM application to generate a response. It's the text or set of instructions given to the LLM application, prompting it to generate output or perform a specific task.
   b. Prompt flow allows you to create flows, which refers to the sequence of actions or steps that are taken to achieve a specific task or functionality. A flow represents the overall process or pipeline that incorporates the interaction with the LLM to address a particular use case.
2. **Understand the development lifecycle of a large language model (LLM) app**
   a. The development lifecycle of a Large Language Model (LLM) application.
      i. Initialization: Define the use case and design the solution.
      ii. Experimentation: Develop a flow and test with a small dataset.
      iii. Evaluation and refinement: Assess the flow with a larger dataset.
      iv. Production: Deploy and monitor the flow and application.
   b. During production, you:
      i. Optimize the flow for efficiency and effectiveness.
      ii. Deploy your flow to an endpoint. When you call the endpoint, the flow is triggered to run and the desired output is generated.
      iii. Monitor the performance of your solution by collecting usage data and end-user feedback. By understanding how the application performs, you can improve the flow whenever necessary.
3. **Understand core components and explore flow types:**
   a. Understand core components and explore flow types:
      i. Prompt flow is a feature within Microsoft Foundry that allows you to author flows. Flows are executable workflows often consist of three parts:
         1. Inputs: Represent data passed into the flow. Can be different data types like strings, integers, or boolean.
         2. Nodes: Represent tools that perform data processing, task execution, or algorithmic operations.
         3. Outputs: Represent the data produced by the flow.
      ii. You can add a node to a flow by choosing one of the available types of tools.
      iii. Explore the tools available in prompt flow:
         1. LLM tool: Enables custom prompt creation utilizing Large Language Models.
         2. Python tool: Allows the execution of custom Python scripts.
         3. Prompt tool: Prepares prompts as strings for complex scenarios or integration with other tools.
      iv. Whenever you add a new node to your flow, adding a new tool, you can define the expected inputs and outputs. A node can use one of the whole flow's inputs, or another node's output, effectively linking nodes together.
      v. If you're looking for functionality that is not offered by the available tools, you can create your own custom tool: https://microsoft.github.io/promptflow/how-to-guides/develop-a-tool/create-and-use-tool-package.html
      vi. There are three different types of flows you can create with prompt flow:
         1. Standard flow: Ideal for general LLM-based application development, offering a range of versatile tools.
         2. Chat flow: Designed for conversational applications, with enhanced support for chat-related functionalities.
         3. Evaluation flow: Focused on performance evaluation, allowing the analysis and improvement of models or applications through feedback on previous runs.
4. **YouTube: AI-900 Azure AI Foundry with Prompt Flow for Named Entity Recognition:**
   a. Prompt flow is a development tool designed to streamline the entire development cycle of AI applications powered by Large Language Models (LLMs).
   b. With Prompt flow, you're able to:
      i. ✓ Create executable flows that link LLMs, prompts, and Python tools through a visualized graph.
      ii. ✓ Debug, share, and iterate your flows with ease through team collaboration.
      iii. ✓ Create prompt variants and evaluate their performance through large-scale testing.
      iv. ✓ Deploy a real-time endpoint that unlocks the full power of your application.



      v. Below is the Prompt Flow UI

vi. 

vii. Below is where you need to put the connection (This will help you connect to your Azure Open AI Services)

viii. 

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5. **Explore connections and runtimes:**
   a. When you create a Large Language Model (LLM) application with prompt flow, you first need to configure any necessary connections and runtimes.
   b. Whenever you want your flow to connect to external data source, service, or API, you need your flow to be authorized to communicate with that external service. When you create a connection, you configure a secure link between prompt flow and external services, ensuring seamless and safe data communication.
   c. Depending on the type of connection you create, the connection securely stores the endpoint, API key, or credentials necessary for prompt flow to communicate with the external service. Any necessary secrets aren't exposed to users, but instead are stored in an Azure Key Vault.
   d. Certain built-in tools require you to have a connection configured:
      i. Connection type : Azure OpenAI | Built-in tools : LLM or Python
      ii. Connection type : OpenAI | Built-in tools : LLM or Python
      iii. Connection type : Azure AI Search | Built-in tools : Vector DB Lookup or Python
      iv. Connection type : Serp | Built-in tools : Serp API or Python
      v. Connection type : Custom | Built-in tools : Python
   e. Prompt flow connections play pivotal roles in two scenarios. They automate API credential management, simplifying and securing the handling of sensitive access information. Additionally, they enable secure data transfer from various sources, crucial for maintaining data integrity and privacy across different environments.
   f. Explore runtimes: After creating your flow, and configuring the necessary connections your tools use, you want to run your flow. To run the flow, you need compute, which is offered through prompt flow runtimes.
      i. Runtimes (1) are a combination of a compute instance (2) providing the necessary compute resources, and an environment (3) specifying the necessary packages and libraries that need to be installed before being able to run the flow.
      ii. When you use runtimes, you have a controlled environment where flows can be run and validated, ensuring that everything works as intended in a stable setting. A default environment is available for quick development and testing. When you require other packages to be installed, you can create a custom environment: https://learn.microsoft.com/en-us/azure/machine-learning/prompt-flow/how-to-customize-session-base-image?view=azureml-api-2
6. **Explore variants and monitoring options:**
   a. During production, you want to optimize and deploy your flow. Finally, you want to monitor your flows to understand when improving your flows is necessary. You can optimize your flow by using variants, you can deploy your flow to an endpoint, and you can monitor your flow by evaluating key metrics.
   b. Prompt flow variants are versions of a tool node with distinct settings. Currently, variants are only supported in the LLM tool, where a variant can represent a different prompt content or connection setting. Variants allow users to customize their approach for specific tasks, like, summarizing news articles.
   c. Some benefits of using variants are:
      i. Enhance the quality of your LLM generation: Creating diverse variants of an LLM node helps find the best prompt and settings for high-quality content.
      ii. Save time and effort: Variants allow for easy management and comparison of different prompt versions, streamlining historical tracking and reducing the effort in prompt tuning.
      iii. Boost productivity: They simplify the optimization of LLM nodes, enabling quicker creation and management of variations, leading to better results in less time.
      iv. Facilitate easy comparison: Variants enable side-by-side result comparisons, aiding in choosing the most effective variant based on data-driven decisions.
   d. When you're satisfied with the performance of your flow, you can choose to deploy it to an online endpoint. Endpoints are URLs that you can call from any application. When you make an API call to an online endpoint, you can expect (almost) immediate response.
   e. When you deploy your flow to an online endpoint, prompt flow generates a URL and key so you can safely integrate your flow with other applications or business processes.
   f. In prompt flow, monitoring evaluation metrics is key to understanding your LLM application's performance, ensuring they meet real-world expectations and deliver accurate results.
7. **ASSESMENT:**

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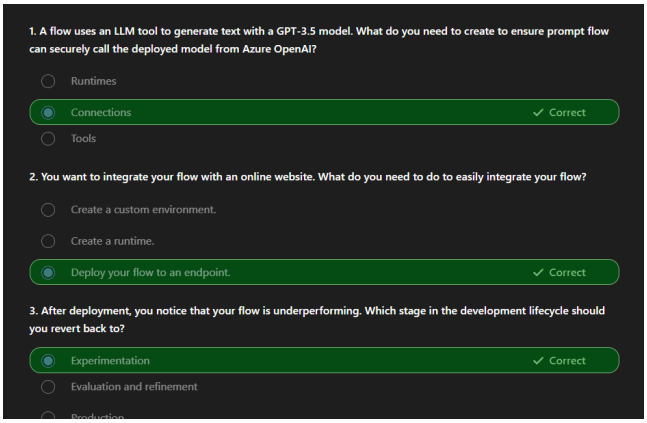

8. Some Extra comments on prompt flow made by users online:
    a. The problem is that prompt flow provides unique capabilities and benefits that simple agents or even A2A do not have currently in Azure Foundry at a low/no code level. For example, the visual nature of a multi-agent flow where there are conditional decisions being made to process a request are unique for both design and sustainability. The simplicity of adding agents or code or just an LLM visually is powerful and simple here.
    b. We have invested many hours of work and training into the integration of prompt flows and not have a path and equal or better technology to transfer these into seamlessly would have significant negative consequences for those of us who invested in this very good approach provided by Microsoft.
    c.